

# **README for Accessing Experimental Real-Time TRMM Multi-Satellite Precipitation Analysis (TMPA-RT) Data Sets**

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## **News Archive**

*17 February 2009* The new Version 6 of the TMPA-RT (not to be confused with the on-going Version 6 of the official TRMM products) was released effective 00Z, Tuesday, 17 February 2009. This is the upgrade that many of you have heard me discuss for over a year, during which time we tested and discarded several alternative schemes. The new version includes the following:

1. The combined precipitation field in 3B42RT is calibrated to the calendar month's climatology of TRMM Version 6 3B42 using local histogram matching of 3-hourly 2B12 (TMI) to 2B31 (TRMM Combined Instrument, or TCI) and ratio adjustment of TCI to 3B43 Version 6. Over land, the major effect is to reduce the bias of 3B42RT, while over ocean it brings 3B42RT into approximate calibration by the 2B31 estimator.
2. To support continued operations with current-style estimates, the uncalibrated combined precipitation field will continue to be provided in 3B42RT, as an additional field at the end of the data file.
3. A data source field is being added at the end of the 3B40RT file, tracking which microwave sensor type provided the input to each grid box; the source field in 3B42RT is being upgraded to pass through the microwave sensor type.
4. MHS-based precipitation estimates from NOAA-18 were quietly added to the then-current TMPA-RT starting on 27 November 2007; and the new version continues to use them.
5. The new release includes the archive of "new" data available starting 00 UTC 01 October 2008 that were processed as part of the beta test. This gives users a backlog of data for starting calibration activities with the new-style data.
6. Upon this release we have deleted all data considered obsolete, namely from versions before 3 February 2005. Data computed with the remaining versions that pre-date Version 6 are moved to subdirectory *old*.
7. Once the new system is stable, MetOp-1 MHS data will be enabled. This enhancement will be announced when it occurs.

## **Introduction**

The system to produce the "TRMM and Other Data" estimates in real time was developed to apply new concepts in merging quasi-global precipitation estimates and to take advantage of the increasing availability of input data sets in near real time. The overall system is referred to as the

real-time TRMM Multi-Satellite Precipitation Analysis (TMPA-RT). The TMPA-RT is run quasi-operationally on a best-effort basis at the TRMM Science Data and Information System (TSDIS), with on-going scientific development by the research team led by Dr. Robert Adler in the GSFC Laboratory for Atmospheres. Estimates are posted to the web about 6 hours after observation time, although processing issues may delay or prevent this schedule. Due to the experimental nature of these estimates, users are encouraged to report their experiences with the data, and they should expect episodic upgrades or outages as the system develops.

## **Product Definitions**

There are three “TRMM and Other Data” products:

### 3B40RT (High Quality, or HQ)

The 3B40RT combined microwave precipitation estimate provides a global 0.25°x0.25°-averaged 3-hourly combination of all available TMI, SSM/I, AMSR-E, and AMSU-B estimates:

1. Offline, the GPROF-SSM/I, GPROF-AMSR, and AMSU-B have been probability-matched to 2A12RT. Prior to 08Z 3 February 2005, this was done routinely for GPROF-SSM/I as 2 sets, one for ocean and one for land, computed every 5 days from the last 30 days of SSM/I-IR match-ups. Thereafter, the calibrations of AMSU-B and AMSR-E to TMI each have one set of coefficients for land and a separate set for ocean, while SSM/I uses one set for land and 5 for ocean, covering the latitude bands 90-30°S, 30-10°S, 10°S-N, 10-30°N, and 30-90°N. AMSR-E uses a 2-month set of match-ups to ensure sufficient sampling, while all of the others work with single-month accumulations. The AMSR-E and AMSU-B coefficients apply to the entire year, while SSM/I uses a separate set for each season.
2. The GPROF-SSM/I, GPROF-AMSR, AMSU-B, and 2A12RT estimates are gridded to a 0.25°x0.25° grid for a 3-hour period centered on the major synoptic times (00Z, 03Z, ..., 21Z). Prior to 08Z 3 February 2005, only GPROF-SSM/I and 2A12RT estimates were used.
3. The GPROF-SSM/I, GPROF-AMSR, and AMSU-B estimates are calibrated to 2A12RT.
4. The rain rate in each grid box is the pixel-weighted average of 2A12RT, GPROF-SSM/I, and GPROF-AMSR grid boxes contributing during the 3 hours, or the pixel-weighted average of AMSU-B if no other HQ estimates are available.
5. The SSM/I and AMSR-E data are available in the latitude band 85°N-S, but GPROF only returns estimates in the band 70°N-S. AMSU-B data are available over the entire globe, and estimates are attempted in the entire domain. However, all of the HQ algorithms are unable to provide estimates in regions with frozen or icy surfaces.
6. In a future upgrade the random error will be estimated. Currently the random error field is set to missing.

### 3B41RT (Variable Rainrate, or VAR)

The 3B41RT IR precipitation estimate converts 0.25°x0.25° -averaged geo-IR Tb to rainrates that are HQ-calibrated locally in time and space:

1. Both geo-IR Tb and HQ are averaged to  $0.25^{\circ} \times 0.25^{\circ}$  to ensure consistent spatial scale, and time-space matched data are accumulated over calendar months.
2. In each calibration, the Tb-rainrate curve is set locally by probability matching the month's histograms of coincident IR Tb and HQ rain rate.

The local VAR Tb-rainrate curve is applied to each Merged 4-Km IR Tb data set in the month:

1. Over most of the globe the on-hour data field is taken as the input data, with fill-in by the previous half-hour image. The exception is the GMS sector, where the previous half-hour is primary, since GMS does not schedule images on the hour. [In that case, much of the GMS sector is filled with data from METEOSAT5 and GOES-W at very high zenith angles.]
2. The Tb-to-rainrate conversion is a simple look-up, using whatever set of VAR calibration coefficients is current.
3. In a future upgrade the random error will be estimated. Currently the random error field is set to missing.

### 3B42RT (Combination of HQ and VAR)

The 3B42RT combination is computed every 3 hours from that hour's HQ and VAR fields:

1. The present combination scheme is to take the HQ field wherever it is non-missing, and fill in with VAR elsewhere.
2. Offline, 10 years of matched 3-hourly TCI and TMI are used to compute monthly calibration histograms, while monthly TCI and 3B43 Version 6 estimates are used to compute monthly calibration ratios. In use, the TMI-TCI and TCI-3B43 calibrations are applied sequentially to the initial 3-hourly HQ+VAR fields.
3. The VAR estimates are only trusted for the latitude band  $50^{\circ}\text{N-S}$ , so both the calibrated and uncalibrated HQ+VAR fields are clipped to  $50^{\circ}\text{N-S}$ , with estimates outside that encoded to negative values. Accordingly, users generally should not employ these values.
4. It is planned to do a more sophisticated combination in a future release.
5. In a future upgrade the random error will be estimated. Currently the random error field is set to missing.

## File Contents and Format

Table 1. File layout for 3B40RT, 3B41RT, 3B42RT.

|       | 3B40RT     |                 | 3B41RT     |          | 3B42RT     |              |
|-------|------------|-----------------|------------|----------|------------|--------------|
| Block | Byte Count | Field           | Byte Count | Field    | Byte Count | Field        |
| 1     | 2880       | header          | 2880       | header   | 2880       | header       |
| 2     | 2073600*   | precip          | 1382400&   | precip   | 1382400&   | precip       |
| 3     | 2073600*   | error           | 1382400&   | error    | 1382400&   | error        |
| 4     | 1036800+   | # pixels        | 691200@    | # pixels | 691200@    | source       |
| 5     | 1036800+   | # ambig. pixels | -          | -        | 1382400&   | uncal precip |
| 6     | 1036800+   | # rain pixels   | -          | -        | -          | -            |
| 7     | 1036800+   | source          |            |          |            |              |

\* INTEGER\*2, 90°N-S                      & INTEGER\*2, 60°N-S

+ INTEGER\*1, 90°N-S                      @ INTEGER\*1, 60°N-S

Header:

Each file starts with a header that is one 2-byte-integer row in length, or 2880 bytes. The header is ASCII in a "PARAMETER=VALUE" format that makes the file self-documenting (e.g., "algorithm\_id=3B40RT"). As such, the header can be read with standard text editors, output as text with simple application programs, or parsed for input into applications. Successive "PARAMETER=VALUE" sets are separated by spaces, and no spaces or "=" are permitted in either PARAMETER or VALUE. The current PARAMETER entries and definitions are:

| PARAMETER                | Definition                                                                        |
|--------------------------|-----------------------------------------------------------------------------------|
| algorithm_ID             | TRMM algorithm identifier (e.g., "3B40RT")                                        |
| algorithm_version        | Version of the science algorithm                                                  |
| granule_ID               | TSDIS granule identifier (e.g., "3B40RT.2001121809.bin")                          |
| header_byte_length       | Number of bytes in the header                                                     |
| file_byte_length         | Number of bytes in the file, expressed as a formula describing the file structure |
| nominal_YYYYMMDD         | Nominal UTC year, month, and day of the month                                     |
| nominal_HHMMSS           | Nominal UTC hour, minute, and second                                              |
| begin_YYYYMMDD           | Start UTC year, month, and day of the month                                       |
| begin_HHMMSS             | Start UTC hour, minute, and second                                                |
| end_YYYYMMDD             | End UTC year, month, and day of the month                                         |
| end_HHMMSS               | End UTC hour, minute, and second                                                  |
| creation_YYYYMMDD        | Date the file was created as year, month, and day of the month                    |
| west_boundary            | Longitude of the western edge of the data domain                                  |
| east_boundary            | Longitude of the eastern edge of the data domain                                  |
| north_boundary           | Latitude of the northern edge of the data domain                                  |
| south_boundary           | Latitude of the southern edge of the data domain                                  |
| origin                   | Geographical direction of the first grid box from the grid center                 |
| number_of_latitude_bins  | Number of grid boxes in the meridional direction                                  |
| number_of_longitude_bins | Number of grid boxes in the zonal direction                                       |
| grid                     | Size of one grid box                                                              |

|                     |                                                                                                      |
|---------------------|------------------------------------------------------------------------------------------------------|
| first_box_center    | Geolocation of the first grid box center                                                             |
| second_box_center   | Geolocation of the second grid box center                                                            |
| last_box_center     | Geolocation of the last grid box center                                                              |
| number_of_variables | Number of data fields                                                                                |
| variable_name       | List of the data field names, separated by commas                                                    |
| variable_units      | List of data field units, separated by commas, in the same order as the variable_name list           |
| variable_scale      | List of data field scaling factors, separated by commas, in the same order as the variable_name list |
| variable_type       | List of data field word types, separated by commas, in the same order as the variable_name list      |
| byte_order          | Order of bytes in a data word ("big_endian" or "little_endian")                                      |
| flag_value          | List of special values, separated by commas                                                          |
| flag_name           | List of special value names, separated by commas, in the same order as the flag_value list           |
| contact_name        | Name of the person to contact with questions                                                         |
| contact_address     | Postal address of the contact_name                                                                   |
| contact_telephone   | Telephone number of the contact_name                                                                 |
| contact_facsimile   | Facsimile number of the contact_name                                                                 |
| contact_email       | Email address of the contact_name                                                                    |

Thereafter the data fields follow. All the fields are on a 0.25° lat./lon. grid that increments most rapidly to the east (from the Prime Meridian) and then to the south (from the northern edge). Grid box edges are on multiples of 0.25°. The data fields are written as flat binary data in big-endian byte order.

### 3B40RT:

Following the header, 6 data fields appear:

|                     |                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| precipitation       | (2-byte integer)                                                                                                                                                                  |
| precipitation_error | (2-byte integer)                                                                                                                                                                  |
| total_pixels        | (1-byte integer)                                                                                                                                                                  |
| ambiguous_pixels    | (1-byte integer; highly uncertain values)                                                                                                                                         |
| rain_pixels         | (1-byte integer)                                                                                                                                                                  |
| source              | (1-byte integer; the values are:<br>0 = no observation    1 = AMSU    2 = TMI    3 = AMSR<br>4 = SSMI            5 = SSMIS    6 = MHS    30 = AMSU&MHS avg.<br>31 = conical avg.) |

All fields are 1440x720 grid boxes (0-360°E,90°N-S). The first grid box center is at (0.125°E,89.875°N). Files are produced every 3 hours on synoptic observation hours (00 UTC, 03 UTC, ..., 21 UTC) as an accumulation of all HQ swath data observed within +/-90 minutes of the nominal file time. Estimates are only computed for the band 70°N-S.

### 3B41RT:

Following the header, 3 data fields appear:

precipitation (2-byte integer)  
precipitation\_error (2-byte integer)  
total\_pixels (1-byte integer)

All fields are 1440x480 grid boxes (0-360°E,60°N-S). The first grid box center is at (0.125°E,59.875°N). Files are produced every hour from the on-hour IR image (except for the previous half-hour image for GMS), with fill-in by the previous half-hour image (except for GMS, where the on-hour image is used for fill-in). Valid estimates are only provided in the band 50°N-S.

### 3B42RT:

Following the header, 4 data fields appear:

precipitation (2-byte integer)  
precipitation\_error (2-byte integer)  
source (1-byte integer; the values are:  
0 = no observation 1 = AMSU 2 = TMI 3 = AMSR  
4 = SSMI 5 = SSMIS 6 = MHS 30 = AMSU&MHS avg.  
31 = conical avg. 50 = IR 1,2,3,4,5,6 + 100 = sparse-sample HQ)  
uncal. precip (2-byte integer)

All fields are 1440x480 grid boxes (0-360°E,60°N-S). The first grid box center is at (0.125°E,59.875°N). Files are produced every 3 hours on synoptic observation hours (00 UTC, 03 UTC, ..., 21 UTC) using that hour's 3B40RT and 3B41RT data sets. Valid estimates are only provided in the band 50°N-S. See "decode high-latitude VAR and HQ+VAR precipitation values" for discussion of retrieving values outside 50°N-S. The leading precipitation field has a climatological bias correction to the 3B42 Version 6 estimates, while the last field is the multi-satellite precipitation before this calibration.

Note that we use the term "gridbox" to denote the values on Level 3 data (i.e., gridded data), while we use the term "pixel" to denote individual values of Level 2 data (i.e., instrument footprints). Thus, there can be many pixels contributing to a gridbox.

Both precipitation and random error are scaled by 100 before conversion to 2-byte integer. Thus, units are 0.01 mm/h. To recover the original floating-point values in mm/h, divide by 100. Missings are given the 2-byte-integer missing value, -31999. The remaining fields are in numbers of pixels, except the source variable, which is dimensionless.

Currently the random error fields are all set to the 2-byte-integer missing value, -31999. This placeholder will be replaced with actual estimates as development proceeds.

The variable ambiguous\_pixels is the count of pixels for which the algorithm cannot determine whether the scene has valid or invalid data. It is a subset of the total\_pixel and many, but not all,

are included in raining\_pixels. In general, a "high" fraction of ambiguous\_pixels indicates that the grid box value is invalid.

The originating machine on which the data files are written is a Silicon Graphics, Inc. Unix workstation, which uses the "big-endian" IEEE 754-1985 representation of 4-byte floating-point unformatted binary numbers. Some CPUs, including PCs and DEC machines, might require a change of representation (i.e., byte swapping) before using the data. In some cases, the gunzip routine, used to uncompress the data, will change representations automatically.

## **Special Values**

All of the scaled 2-byte-integer precipitation and random error fields have one value with special meaning. Any grid box with insufficient valid data to make an estimate is assigned the 2-byte-integer value -31999. As well, the scaled 2-byte-integer precipitation and random error fields are clipped to [-31998,31998] to prevent duplication of the missing value (at the negative end) or overflows (at both the positive and negative ends). Note that any examples of clipping should be immediately reported to the dataset developers.

In addition, 3B40RT precipitation values that are highly likely to be artifacts (ambiguous fraction of pixels at least 40% or 5x5-gridbox averaged ambiguous fraction at least 20%) are encoded as  $(-p - 0.01)$ , where "p" is the original precipitation value, before conversion to scaled 2-byte-integer. Thus, users can recover the estimated value of such gridboxes if desired, but the usual scheme of requiring precipitation to be non-negative will filter out these suspect values.

Likewise, 3B41RT and 3B42RT precipitation values outside the 50°N-S latitude band are considered experimental and are encoded as  $(-p - 0.01)$ , where "p" is the original precipitation value, before conversion to scaled 2-byte-integer. Thus, users can recover the estimated value of such gridboxes if desired, but the usual scheme of requiring precipitation to be non-negative will filter out these suspect values.

The 3B42RT "source of estimate" field only has three discrete values, -1, 0, 100, which correspond to "no estimate", "HQ", and "VAR".

Note that any negative values in the various "number of" fields is a processing error that should be immediately reported to the dataset developers.

## **Dataset Validation**

Formal validation studies are underway. The time series of the global images shows good continuity in time and space across the geo-IR data boundaries. Overall, the analysis approach appears to be working as expected. See Huffman et al. (2007, 2009) for more information.

Validation studies are being conducted under the auspices of the International Precipitation Working Group in Australia, the continental U.S., western Europe, and parts of South America. Respectively, the web sites for these activities are:

[http://www.bom.gov.au/bmrc/SatRainVal/sat\\_val\\_australia.html](http://www.bom.gov.au/bmrc/SatRainVal/sat_val_australia.html)  
[http://www.cpc.ncep.noaa.gov/products/janowiak/us\\_web.shtml](http://www.cpc.ncep.noaa.gov/products/janowiak/us_web.shtml)  
<http://kermi.bham.ac.uk/~ipwgeu/>  
<http://cics.umd.edu/~dvila/web/SatRainVal/dailyval.html>

The primary limitations on the HQ (3B40RT) are the sparse sampling by the collection of passive-microwave satellites and algorithm drop-outs in regions with icy or frozen surface. The infrared results (3B41RT) are designed to emulate the microwave results as closely as possible, so known deficiencies in the microwave will likely be reflected in the infrared as well. In addition, it is well-known that infrared algorithms of the kind used here have large random errors at the fine time and space scales provided. However, we expect the infrared estimates to match the histogram of microwave estimates, so that user-specified averaging should yield approximately unbiased results. Finally, the combined microwave-IR fields (3B42RT) contain data boundaries between the regions of microwave and IR coverage. Instantaneously the boundaries are usually subtle, but are more noticeable in movie loops, since the regions of coverage change with each image. We encourage users to report successes and problems in applying these datasets to their particular applications.

## **Dataset Status**

Beta testing began in early December 2001. An official (experimental) version was instituted in late January 2002. Processing changes occurred on 6 February and 12 March 2002, and subsequent developments are documented in the “Latest News” and “Previous News” at the top of this document. Users should anticipate a series of versions as the algorithm is developed further. The present areas of interest are: improving the HQ product by auditing out AMSU-B data that are deficient in precipitation coverage; moving to shorter-interval estimation periods to more accurately represent the time series of precipitation; and adding data at high latitudes.

Currently, access to the digital data is by anonymous FTP to [trmmopen.gsfc.nasa.gov](ftp://trmmopen.gsfc.nasa.gov). Under subdirectory `pub/merged` there are three directories:

- CombinedMicro (3B40RT)
- CalibratedIR (3B41RT)
- MergeIRMicro (3B42RT)

## **Example Programs**

The data fields are all written with C-language code as blocks of bytes, so there are no extraneous bytes in the files. Because the first two fields are 2-byte integers and the rest are 1-byte integers in each file (to save space), users must exercise care in using FORTRAN direct access to read the data. The FORTRAN example programs read all fields with a single OPEN.

Alternatively, the files can be opened with different logical record sizes depending on whether one is reading 2-byte-integer or 1-byte-integer fields. Note well that the units of the logical record size is not part of the FORTRAN 77 standard. On SGI machines it is in 4-byte words, but some other systems expect it in bytes. Also, to repeat an earlier comment, the originating machine on which the data files are written is a Silicon Graphics, Inc. Unix workstation. It uses the "big-endian" IEEE 754-1985 representation of 4-byte floating-point unformatted binary numbers, and some CPUs, such as PCs and DEC machines, might require a change of representation (i.e., byte swapping) before using the data.

The FTP site <ftp://trmmopen.gsfc.nasa.gov/pub/merged/software> provides several example programs:

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| idlsave.pro      | IDL journal file showing output of a particular 3B40RT header |
| read3B4XRT.c     | C example                                                     |
| read_header.f    | FORTTRAN header-read example                                  |
| read_rt_file.f   | FORTTRAN single-read example                                  |
| read_rt_file.pro | IDL example                                                   |
| read_rt_lines.f  | FORTTRAN line-by-line example                                 |

These are also available at [ftp://meso.gsfc.nasa.gov/pub/agnes/huffman/rt\\_examples/rt\\_docs](ftp://meso.gsfc.nasa.gov/pub/agnes/huffman/rt_examples/rt_docs).

## Example Images and Movies

Users may obtain example GIF images and QuickTime movies at <http://trmm.gsfc.nasa.gov>. In addition, users may create their own subsets, averages, and time series, downloadable as plots or ASCII data from the TRMM Online Visualization System (TOVAS), at <http://disc2.nascom.nasa.gov/Giovanni/tovas/>. TOVAs is provided by the Goddard Earth Science Data and Information Services Center (GES DISC).

## Additional Documentation

Users should refer to the detailed documentation (3B4XRT\_doc.pdf) and programming examples at the anonymous FTP site <ftp://trmmopen.gsfc.nasa.gov/pub/merged/software> for additional details.

## Primary References

Huffman, G.J., R.F. Adler, D.T. Bolvin, G. Gu, E.J. Nelkin, K.P. Bowman, Y. Hong, E.F. Stocker, D.B. Wolff, 2007: The TRMM Multi-satellite Precipitation Analysis: Quasi-global, multi-year, combined-sensor precipitation estimates at fine scale. *J. Hydrometeor.*, **8**(1), 38-55. PDF available at [ftp://meso.gsfc.nasa.gov/agnes/huffman/papers/TMPA\\_jhm\\_07.pdf.gz](ftp://meso.gsfc.nasa.gov/agnes/huffman/papers/TMPA_jhm_07.pdf.gz)

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